

CS 323: Final Project - Legend

< prog >	:=	P
< identifier >	:=	I
< dec-list >	:=	B
< dec >	:=	C
< type >	:=	K
< stat-list >	:=	G
< stat >	:=	H
< write >	:=	W
< str >	:=	J
< assign >	:=	A
< expr >	:=	E
< term >	:=	T
< factor >	:=	F
< number >	:=	N
< sign >	:=	S
< digit >	:=	D
< letter >	:=	L

< expr >'	:=	Q
< term >'	:=	R
intermediate 1	:=	X
intermediate 2	:=	Y

~~Non-terminals:~~ P I

Terminals: {program ; var begin end λ : , integer "value=";
 = + - * / () * print 0 1 2
 3 4 5 6 7 8 9 a b c d l f }

$$I \rightarrow L \{ \underline{L|D} \}$$

X

$$\begin{aligned} I &\rightarrow LX \\ X &\rightarrow LX \\ X &\rightarrow DX \\ X &\rightarrow \lambda \end{aligned}$$

$$N \rightarrow SD \{ \underline{D} \}$$

Y

$$\begin{aligned} N &\rightarrow SDY \\ Y &\rightarrow DY \\ Y &\rightarrow \lambda \end{aligned}$$

$$\begin{aligned} E &\rightarrow E + T \quad \alpha \\ E &\rightarrow E - T \quad \alpha \\ E &\rightarrow T \quad \beta \end{aligned}$$

$$\begin{aligned} E &\rightarrow TQ \\ Q &\rightarrow +TQ \\ Q &\rightarrow -TQ \\ Q &\rightarrow \lambda \end{aligned}$$

$$\begin{aligned} T &\rightarrow T * F \quad \alpha \\ T &\rightarrow T / F \quad \alpha \\ T &\rightarrow F \quad \beta \end{aligned}$$

$$\begin{aligned} T &\rightarrow FR \\ R &\rightarrow *FR \\ R &\rightarrow /FR \\ R &\rightarrow \lambda \end{aligned}$$

CS 323: Final Project - Complete CFG in BNF

$P \rightarrow \text{program } I ; \text{ var } B \text{ begin } G \text{ end}$

$I \rightarrow L X$

$X \rightarrow L X$

$X \rightarrow D X$

$X \rightarrow \lambda$

$B \rightarrow C : ~~K~~ ;$

$C \rightarrow I , C$

$C \rightarrow I$

$K \rightarrow \text{integer}$

$G \rightarrow H$

$G \rightarrow H G$

$H \rightarrow W$

$H \rightarrow A$

$W \rightarrow \text{print } (J I)$

$J \rightarrow \text{"value="};$

$J \rightarrow \lambda$

$A \rightarrow I = E ;$

$E \rightarrow T Q$

$Q \rightarrow + T Q$

$Q \rightarrow - T Q$

$Q \rightarrow \lambda$

$T \rightarrow F R$

$R \rightarrow * F R$

$R \rightarrow / F R$

$R \rightarrow \lambda$

$F \rightarrow I$

$F \rightarrow N$

$F \rightarrow (E)$

$N \rightarrow S D Y$

$Y \rightarrow D Y$

$Y \rightarrow \lambda$

$S \rightarrow +$

$S \rightarrow -$

$S \rightarrow \lambda$

$D \rightarrow 0$

$D \rightarrow 1$

$D \rightarrow 2$

$D \rightarrow 3$

$D \rightarrow 4$

$D \rightarrow 5$

$D \rightarrow 6$

$D \rightarrow 7$

$D \rightarrow 8$

$D \rightarrow 9$

$L \rightarrow a$

$L \rightarrow b$

$L \rightarrow c$

$L \rightarrow d$

$L \rightarrow l$

$L \rightarrow f$

$$\text{FIRST}(L) = \{a \ b \ c \ d \ \backslash f\}$$

$$\text{FIRST}(D) = \{0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9\}$$

$$\text{FIRST}(S) = \{+ \ - \ \backslash\}$$

$$\text{FIRST}(Y) = \{\backslash \ 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9\}$$

$$\text{FIRST}(N) = \{+ \ - \ \backslash \ 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9\}$$

$$\text{FIRST}(F) = \{(\ + \ - \ 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ a \ b \ c \ d \ \backslash f\}$$

$$\text{FIRST}(R) = \{* \ / \ \backslash\}$$

$$\text{FIRST}(T) = \{(\ + \ - \ 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ a \ b \ c \ d \ \backslash f\}$$

$$\text{FIRST}(Q) = \{+ \ - \ \backslash\}$$

$$\text{FIRST}(E) = \{(\ + \ - \ 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ a \ b \ c \ d \ \backslash f\}$$

$$\text{FIRST}(A) = \{a \ b \ c \ d \ \backslash f\}$$

$$\text{FIRST}(J) = \{\backslash \ \text{"value="} \}$$

$$\text{FIRST}(W) = \{\text{print}\}$$

$$\text{FIRST}(H) = \{\text{print} \ a \ b \ c \ d \ \backslash f\}$$

$$\text{FIRST}(G) = \{\text{print} \ a \ b \ c \ d \ \backslash f\}$$

$$\text{FIRST}(K) = \{\text{integer}\}$$

$$\text{FIRST}(C) = \{a \ b \ c \ d \ \backslash f\}$$

$$\text{FIRST}(B) = \{a \ b \ c \ d \ \backslash f\}$$

$$\text{FIRST}(X) = \{\backslash \ 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8 \ 9 \ a \ b \ c \ d \ \backslash f\}$$

$$\text{FIRST}(I) = \{a \ b \ c \ d \ \backslash f\}$$

$$\text{FIRST}(P) = \{\text{program}\}$$

CS 323: Final Project - Members of FOLLOW

$$\text{FOLLOW}(P) = \{ \$ \}$$

P is starting $\therefore \$ \in \text{FOLLOW}(P)$

$$\text{FOLLOW}(I) = \{ ; ,) \}$$

$P \rightarrow \text{program } I; \text{ var } B \text{ begin } G \text{ end}$ $\therefore ; \in \text{FOLLOW}(I)$

$C \rightarrow I, C$ $\therefore , \in \text{FOLLOW}(I)$

$W \rightarrow \text{print}(JI)$ $\therefore) \in \text{FOLLOW}(I)$

$A \rightarrow I = E;$ $\therefore = \in \text{FOLLOW}(I)$

$$\text{FOLLOW}(X) = \{ ; ,) \}$$

$I \rightarrow LX$ $\therefore \text{FOLLOW}(I) \subseteq \text{FOLLOW}(X)$

$X \rightarrow LX$ No New info

$X \rightarrow DX$ no new info

$X \rightarrow \lambda$ no new info

$$\text{FOLLOW}(B) = \{ \text{begin} \}$$

$P \rightarrow \text{program } I; \text{ var } B \text{ begin } G \text{ end}$ $\therefore \text{begin} \in \text{FOLLOW}(B)$

$$\text{FOLLOW}(C) = \{ : \}$$

$B \rightarrow C : K;$ $\therefore : \in \text{FOLLOW}(C)$

$$\text{FOLLOW}(K) = \{ ; \}$$

$B \rightarrow C : K;$ $\therefore ; \in \text{FOLLOW}(K)$

$$\text{FOLLOW}(G) = \{ \text{end} \}$$

$P \rightarrow \text{program } I; \text{ var } B \text{ begin } G \text{ end}$ $\therefore \text{end} \in \text{FOLLOW}(G)$

$G \rightarrow HG$ No new info

$$\text{FOLLOW}(H) = \{ \text{end print a b c d l f} \}$$

$G \rightarrow H$ $\therefore \text{FOLLOW}(G) \subseteq \text{FOLLOW}(H)$

$G \rightarrow HG$ $\therefore \text{FIRST}(G) \subseteq \text{FOLLOW}(H)$

$$\text{FOLLOW}(W) = \{ \text{end print a b c d l f} \}$$

$H \rightarrow W$ $\therefore \text{FOLLOW}(H) \subseteq \text{FOLLOW}(W)$

$$\text{FOLLOW}(J) = \{ \text{a b c d l f} \}$$

$W \rightarrow \text{print}(JI)$ $\therefore \text{FIRST}(J) \subseteq \text{FOLLOW}(J)$

$$\text{FOLLOW}(A) = \{ \text{end print a b c d l f} \}$$

$H \rightarrow A$ $\therefore \text{FOLLOW}(H) \subseteq \text{FOLLOW}(A)$

$$\text{FOLLOW}(E) = \{ ;) \}$$

$A \rightarrow I = E;$ $\therefore ; \in \text{FOLLOW}(E)$

$F \rightarrow (E)$ $\therefore) \in \text{FOLLOW}(E)$

$$\text{FOLLOW}(Q) = \{ ;) \}$$

$E \rightarrow TQ$	$\therefore \text{FOLLOW}(E) \subseteq \text{FOLLOW}(Q)$
$Q \rightarrow +TQ$	no new info
$Q \rightarrow -TQ$	no new info
$Q \rightarrow \lambda$	no new info

$$\text{FOLLOW}(T) = \{ + - \times ;) \}$$

$E \rightarrow TQ$	$\therefore \text{FIRST}(Q) \subseteq \text{FOLLOW}(T)$
$Q \rightarrow \lambda \Rightarrow E \rightarrow T$	$\therefore \text{FOLLOW}(E) \subseteq \text{FOLLOW}(T)$
$Q \rightarrow +TQ$	$\therefore \text{FOLLOW}(Q) \subseteq \text{FOLLOW}(T)$
$Q \rightarrow \lambda$	
$Q \rightarrow -TQ$	no new info

$$\text{FOLLOW}(R) = \{ + - ;) \}$$

$T \rightarrow FR$	$\therefore \text{FOLLOW}(T) \subseteq \text{FOLLOW}(R)$
$R \rightarrow *FR$	no new info
$R \rightarrow /FR$	no new info
$R \rightarrow \lambda$	no new info

$$\text{FOLLOW}(F) = \{ * / \times + - ;) \}$$

$T \rightarrow FR$	$\therefore \text{FIRST}(R) \subseteq \text{FOLLOW}(F)$
$R \rightarrow \lambda \Rightarrow T \rightarrow F$	$\therefore \text{FOLLOW}(T) \subseteq \text{FOLLOW}(F)$
$R \rightarrow *FR$	$\therefore \text{FOLLOW}(R) \subseteq \text{FOLLOW}(F)$
$R \rightarrow \lambda$	
$R \rightarrow /FR$	no new info

$$\text{FOLLOW}(N) = \{ * / + - ;) \}$$

$F \rightarrow N$	$\therefore \text{FOLLOW}(F) \subseteq \text{FOLLOW}(N)$
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$$\text{FOLLOW}(Y) = \{ * / + - ;) \}$$

$N \rightarrow SDY$	$\therefore \text{FOLLOW}(N) \subseteq \text{FOLLOW}(Y)$
$Y \rightarrow DY$	no new info
$Y \rightarrow \lambda$	no new info

$$\text{FOLLOW}(S) = \{ 0 1 2 3 4 5 6 7 8 9 \}$$

$N \rightarrow SDY$	$\therefore \text{FIRST}(D) \subseteq \text{FOLLOW}(S)$
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$$\text{FOLLOW}(D) = \{ 0 1 2 3 4 5 6 7 8 9 \times * / + - ;) , = \}$$

$N \rightarrow SDY$	$\therefore \text{FIRST}(Y) \subseteq \text{FOLLOW}(D)$
$Y \rightarrow \lambda \Rightarrow N \rightarrow SD$	$\therefore \text{FOLLOW}(N) \subseteq \text{FOLLOW}(D)$
$X \rightarrow DX$	$\therefore \text{FOLLOW}(X) \subseteq \text{FOLLOW}(D)$
$X \rightarrow \lambda$	
$N \rightarrow SDY$	
$Y \rightarrow \lambda$	$\therefore$ No new info
$S \rightarrow \lambda$	

$$\text{FOLLOW}(L) = \{0 1 2 3 4 5 6 7 8 9 a b c d l f \# ; ,) = \}$$

$$\begin{array}{l} I \rightarrow LX \\ X \rightarrow \lambda \\ X \rightarrow LX \\ X \rightarrow \lambda \end{array} \left. \vphantom{\begin{array}{l} I \rightarrow LX \\ X \rightarrow \lambda \\ X \rightarrow LX \\ X \rightarrow \lambda \end{array}} \right\} \begin{array}{l} I \rightarrow L \\ X \rightarrow L \end{array}$$

$$\begin{array}{l} \therefore \text{FIRST}(X) \subseteq \text{FOLLOW}(L) \\ \therefore \text{FOLLOW}(I) \subseteq \text{FOLLOW}(L) \\ \therefore \text{FOLLOW}(X) \subseteq \text{FOLLOW}(L) \end{array}$$